RAJALAKSHMI ENGINEERING COLLEGE

**Rakshith Y G 250701578-2025-CSE** R2

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|---|---|
| **Started on** | Monday, 3 August 2026, 11:35 AM |
| **State** | Finished |
| **Completed on** | Friday, 7 August 2026, 4:19 PM |
| **Time taken** | 4 days 4 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

void func(int n)
{
    int counter = 0;

    if(n == 1)
    {
        // printf("*");    // remove printing
        counter++;    // count printf
    }
    else
    {
        counter++; // initial entry into else

        for(int i = 1; i <= n; i++)
        {
            counter++; // outer loop comparison

            for(int j = 1; j <= n; j++)
            {
                counter++; // inner loop comparison

                // printf("*");  // remove printing
                counter++; // count printf

                // printf("*");  // remove printing
                counter++; // count printf

                counter++; // count break
                break;
            }
        }

        counter++; // final outer loop failed comparison
```

```
    }

    printf("%d", counter);
}

int main()
{
    int n;
    scanf("%d", &n);
    func(n);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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